Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/16/2019
PI Name:	Chen, Lung-Chi Ph.D.		
Project Title:	Adverse Cardiopulmonary Effects of Exposure to Extraterrestrial Dusts - Implications for Space Travel		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10987-5007	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Crew Health (FLAGSHIP, NSBRI, OMNIBUS). Appendix A-Crew Health, Appendix B-NSBRI, Appendix C-Omnibus
Start Date:	08/08/2016	**End Date:**	08/07/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Lemery, Jay	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/7/2018 per NSSC information (Ed., 5/10/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Harrington, Andrea Ph.D. (New York University School of Medicine)		
Grant/Contract No.:	NNX16AR03G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The risks to human health associated with exposure to celestial dust during space exploration are largely unknown. Our preliminary research demonstrates that there are significant differences in the pulmonary inflammation generated upon acute exposure to six celestrial materials originating from the Moon, Mars, and asteroid 4Vesta. More specifically, the acute exposure studies reveal interesting relationships between toxicity and a meteorite sample's origin, its pre-ejected state (unweathered versus processed), and geochemical features. The proposed research will build upon this preliminary investigation. The addition of more extraterrestrial samples, chronic health effects studies, and cardiovascular endpoints will: 1) help elucidate the hazards associated with exposure to celestial dust and 2) provide a more robust dataset for the development of a model that utilizes geochemical features (i.e., elemental composition, mineralogy, surface area, and acellular reactivity) to generate a preliminary risk assessment before touchdown. The proposed ground-based research project addresses all of the gaps listed by the Human Research Program (HRP) Human Research Roadmap within the Risk of Adverse Health and Performance Effects of Celestial Dust Exposure initiative. An array of techniques within the fields of environmental toxicology and geosciences are employed allowing for both focused and broad conclusions to be drawn. This approach is highly interdisciplinary and relies on vetted techniques that have been combined in novel ways. In summary, the proposed research will not only identify possible inhalation hazards likely encountered in future space travel, but will allow for the development of a geochemical model to help mitigate the risk of celestial dust exposure. We hypothesize that geochemical differences between the samples will affect the toxicity of a material and that the deleterious nature of a celestial dust will be able to be predicted upon completion of this study.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research has identified possible inhalation hazards likely encountered in future space travel. We have developed a geochemical model to help mitigate the risk of celestial dust exposure.
Task Progress:	This ground-based research project addresses four specific gaps in knowledge identified by the Human Research Roadmap utilizing an array of analytical techniques in environmental toxicology and geosciences. By building upon a multi-faceted investigation of the acute toxicity of meteorite samples originating from the Moon, Mars, and asteroid 4Vesta, it is possible to derive both narrow and broad conclusions. Our preliminary research demonstrated that there are significant differences in the pulmonary inflammation generated upon acute exposure to the six celestrial materials utilized, as well as the mid-ocean ridge basalt. More specifically, the acute exposure studies reveal interesting relationships between toxicity and a meteorite sample's origin, its pre-ejected state (unweathered versus processed), and geochemical features. The present study builds upon this preliminary investigation by increasing the exposure duration and experimental endpoints. We hypothesized that geochemical differences between the samples, however small, will affect the toxicity of a material and that a geochemical model may be developed to help understand and mitigate the health risk to the astronauts. While the former hypothesis has been verified, proving the later hypothesis requires further study. One of the main results of the present study is that geochemical and toxicological correlations made during acute studies do not hold for subacute studies; however, other correlations developed. For example, there is no correlation with bulk iron content and neutrophil (or PMN) infiltration into the lungs in subacute studies as there was for acute exposure but there is a correlation with particle derived reactive oxygen species (ROS) formation and PMN infiltration. Given the role that ROS have in the body's inflammatory response, it is not unexpected that this correlation would develop over time. In fact, one of the possible explanations for why there was no acute correlation with particle derived ROS is that there was not enough time for the particulate generated ROS to overwhelm the initial biological inflammatory response to the particles. Although no longer correlated with PMN infiltration, there is a correlation between bulk chemical composition and cardiopulmonary endpoints. Specifically, there are strong correlations between: 1) IL-1ß dysregulation and chromium and manganese content after a three day exposure (both positive), 2) IL-1ß dysregulation and chromium and titanium content after a three week exposure (positive and negative, respectively), and 3) TNF-a dysregulation and titanium content (negative). There is also a very strong correlation indicating that increasing titanium concentrations enhance vasoconstriction. In conclusion, this study demonstrates the importance of geochemical features in generating inflammation and that there are significant differences between dusts from within the sample planetary body. Although more research is required, the use of geochemical features to help preemptively assess human health hazards may still be viable but likely require ground-based wet chemistry and not just bulk chemical analyses collected in orbit or on the surface.
Bibliography Type:	Description: (Last Updated: 05/21/2019)
Abstracts for Journals and Proceedings	Harrington AD, McCubbin FM, Kaur J, Smirnov A, Galdanes K, Schoonen MAA, Chen LC, Tsirka SE, Gordon T. "Pulmonary Inflammatory Responses to Acute Meteorite Dust Exposures – Implications for Human Space Exploration." Presented at 48th Lunar and Planetary Science Conference (LPSC), The Woodlands, Texas, March 20–24, 2017. 48th Lunar and Planetary Science Conference (LPSC), The Woodlands, Texas, March 20–24, 2017. Abstracts. , Mar-2017
Abstracts for Journals and Proceedings	Harrington AD, McCubbin FM, Kaur J, Smirnov A, Galdanes K, Schoonen MAA, Chen LC, Tsirka SE, Gordon T. "Acute Meteorite Dust Exposure and Pulmonary Inflammation – Implications for Human Space Exploration." Presented at Dust in the Atmosphere of Mars and Its Impact of Human Exploration, Houston, TX, June 13-15, 2017. Abstracts. Dust in the Atmosphere of Mars and Its Impact of Human Exploration, Houston, TX, June 13-15, 2017. , Jun-2017
Abstracts for Journals and Proceedings	Harrington AD, McCubbin FM, Kaur J, Smirnov A, Galdanes K, Schoonen MAA, Chen LC, Tsirka SE, Gordon T. "Pulmonary Inflammatory Responses to Acute Meteorite Dust Exposures – Implications for Human Space Exploration." Goldschmidt2017 conference, Paris, France, August 13-18, 2017. Abstracts. Goldschmidt2017 conference, Paris, France, August 13-18, 2017. , Aug-2017

Abstracts for Journals and Proceedings	Vander Kaaden KE, Harrington AD, McCubbin FM. "Meteorite Dust and Health – A Novel Approach for Determining Bulk Compositions for Toxicological Assessments of Precious Materials." Consortium for Materials Properties Research in Earth Science (COMPRES), Albuquerque, NM, July 9-12, 2017. Consortium for Materials Properties Research in Earth Science (COMPRES), Albuquerque, NM, July 9-12, 2017. Abstract may be found: https://ntrs.nasa.gov/search.jsp?R=20170005626 , Jul-2017
Abstracts for Journals and Proceedings	Harrington AD, McCubbin FM, Kaur J, Smirnov A, Galdanes K, Schoonen MAA, Chen LC, Tsirka SE, Gordon T. "Pulmonary Inflammatory Responses to Acute Meteorite Dust Exposures – Implications for Human Space Exploration." 3rd International Conference of Atmospheric Dust, Bari, Italy, May 29-31, 2018. Abstracts. 3rd International Conference of Atmospheric Dust, Bari, Italy, May 29-31, 2018. , May-2018